



FINAL YEAR PROJECT

Clinic Queue & Patient Flow Management System

A smart, web-based solution for university clinic patient registration, prioritization, and real-time queue management.

University clinics are overwhelmed.

Long queues, missed emergencies, and paper-based systems create a broken patient experience — costing time and, in critical cases, lives.



Overcrowding & Long Waits

No visibility into queue length or wait time



Missed Emergencies

Critical patients not triaged fast enough



Paper-Based Inefficiency

Manual records cause errors and slow throughput

A web-based smart queue system

Built for real university clinic conditions — no app install required, no hardware needed.



Smart Registration

Patients register via browser with symptom input and instant queue assignment



Priority Engine

Automated triage ranks patients by emergency level, symptoms, and age



Live Dashboard

Staff get a real-time queue board and reporting tools at a glance

KEY FEATURES



Smart Registration

Web-based patient intake with symptom capture and real-time queue placement



Emergency Detection

Symptom-based triage algorithm with age factor for automatic priority scoring



Queue Display Board

Live public screen showing current queue state — zero staff intervention needed



Reporting Dashboard

Admin analytics: daily volumes, wait times, peak hours, and case history

How emergencies are caught instantly

01

Symptom Input

Patient selects symptoms during registration — chest pain, difficulty breathing, etc.

02

Logic Scoring

Each symptom carries a severity weight; combined score determines urgency tier

03

Age Prioritization

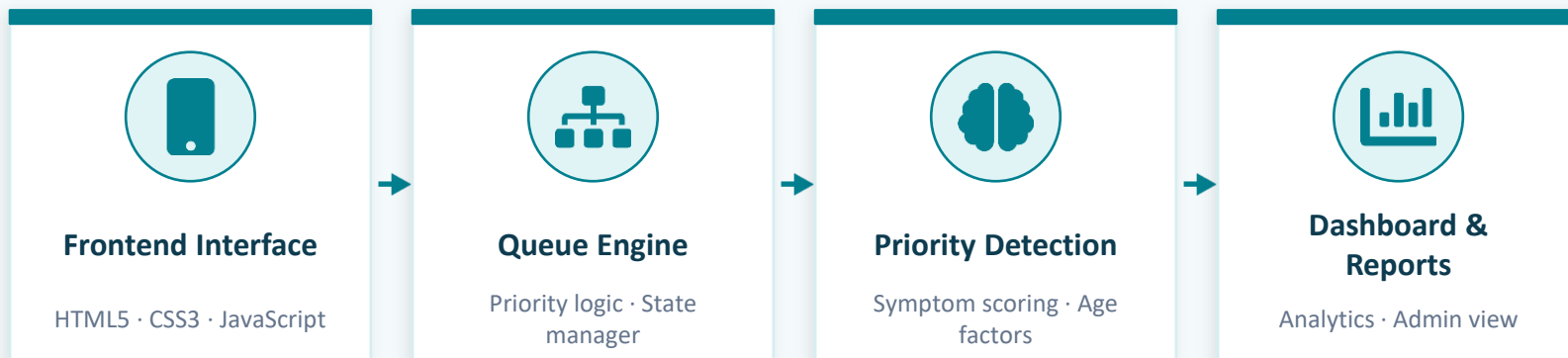
Elderly (60+) and infant patients automatically receive elevated priority tier

04

Queue Positioning

High-priority patients jump the queue; staff dashboard flags them in red

A clean, layered architecture for reliability and speed



Each layer is independently testable — the system degrades gracefully if any component is unavailable.

Built lean — runs everywhere



HTML5

Semantic structure & accessibility



CSS3

Responsive layouts & animations



JavaScript

Queue logic, DOM control & state



No Backend

Zero server dependency for v1.0



No Install

Runs in any modern browser



Scalable

Designed for DB integration in v2.0

IMPACT & BENEFITS



60%

Faster triage

Automated priority detection vs manual assessment



100%

Emergency capture rate

No patient skipped — every admission is scored



Zero

Hardware required

Runs on any browser — phones, tablets, or PCs

Where we go from here

Phase 2



Database Integration

Persistent MySQL/PostgreSQL backend with patient history and analytics export

Phase 3



Cloud Deployment

Host on AWS or Firebase for multi-clinic, institution-wide access

Phase 4



SMS Notifications

Patients receive queue updates via SMS — no need to wait inside the clinic

Phase 5



Mobile App (Flutter)

Native mobile app version for patients and staff with offline capability



Thank You

Clinic Queue & Patient Flow Management System

Questions & Discussion Welcome

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